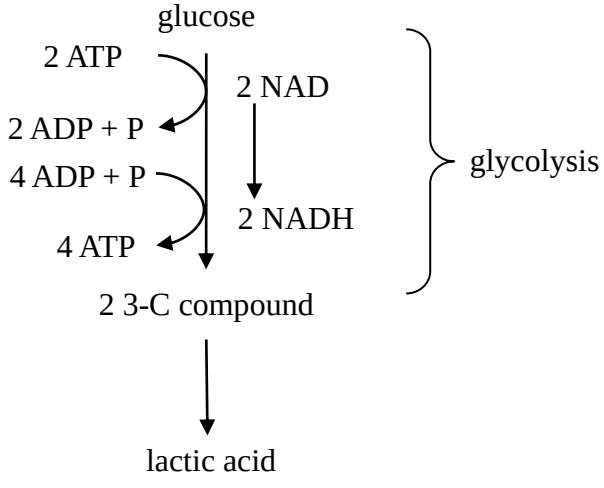
Short questions

|!|BCESB0220000001|!| ★

The diagram below shows the process of anaerobic respiration in muscles cells during exercise. Lactic acid is formed during the process.



(a) Explain why glycolysis initially uses two ATP molecules before producing more ATP molecules. (1 mark)

(b) State ***one*** advantage of using ATP as the energy-storage molecule in a cell. (1 mark)

(c) (i) Name the 3-C compound. (1 mark)

(ii) Describe how lactic acid is produced from the 3-C compound. (2 marks)

(d) The accumulation of lactic acid is toxic to the human body. Describe how lactic acid and its derivative are processed by the body after exercise. (2 marks)

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##

(a) Glucose is a stable molecule. ATP provides the energy needed to activate it. (1)

(b) Any ***one*** of the following: (1)

- ATP cannot pass out of cells
- ATP can be readily broken down to release energy
- Each ATP molecule stores / releases a small amount of energy

(c) (i) Pyruvate (1)

(ii) NADH transfers its hydrogen to pyruvate. (1)

Pyruvate is therefore reduced to form lactic acid. (1)

(d) Lactic acid is transported to the liver and converted back to pyruvate. (1)

Pyruvate enters the Krebs cycle and is oxidized to release more energy.

OR

Pyruvate is converted to glycogen and stored in the liver. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)(i)	Analysing data	1
(c)(ii)	Understanding basic concepts	2
(d)	Understanding basic concepts	2

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|!|BCESB0220000002|!| ★

For each of the biomolecules listed in column 1, select from column 2 ***one*** phrase that matches it. Put the appropriate letter in the space provided. (3 marks)

Column 1			***Column 2***
NAD	(a)	_____	A. a product of photochemical reactions
pyruvate	(b)	_____	B. a product of the Calvin cycle
NADPH	(c)	_____	C. a product of glycolysis
			D. a product of the Krebs cycle
			E. a product of oxidative phosphorylation

##

(a) E (1)
(b) C (1)
(c) A (1)

(3 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1

|!|BCESC0220000003|!|

The following paragraph describes anaerobic respiration in yeast. Complete the paragraph with the appropriate word(s). (5 marks)

Anaerobic respiration in yeast is called (a) _____ fermentation. In the process called (b) _____, glucose is first converted to 3-C compounds called (c) _____, which causes the synthesis of ATP. When there is insufficient oxygen, the 3-C compound loses a carbon dioxide molecule and accepts hydrogen from (d) _____ to form (e) _____.

##

(a) alcoholic (1)
(b) glycolysis (1)
(c) pyruvate (1)
(d) NADH (1)
(e) ethanol (1)

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(5 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

|!|BCESA0220000004|!| ★★

The following photograph shows an eastern emerald elysia sea slug (*Elysia chlorotica*) with its soft leaf-shaped body. It obtains chloroplasts from the algae it eats, and stores them in the cells that line its digestive tract. Young individuals fed with algae for two weeks can survive for the rest of their lives without eating.



Source: https://www.flickr.com/photos/117184384@N07/13048556073
Credit: Natalia Agudelo, Smithsonian National Museum of Natural History

(a) Is the eastern emerald elysia an autotroph? Explain your answer. (1 mark)

_____

_____

(b) The main product of photosynthesis provides energy to the eastern emerald elysia and allows the sea slug to survive for long periods of time without eating. Name this product of photosynthesis. (1 mark)

_____

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(c) The eastern emerald elysia can gain energy from the product of photosynthesis through respiration

(i) Name ***two*** stages of respiration that can produce ATP. (2 marks)

(ii) Briefly describe how the stages in (i) can produce ATP. (2 marks)

(d) Aquatic environments have low concentrations of dissolved oxygen. Explain how the presence of chloroplasts allows the eastern emerald elysia to overcome this disadvantage. (1 mark)

##

(a) No, because it must first consume algae in order to obtain the chloroplasts. (1)

(b) Glucose (1)

(c) (i) Any ***two*** of the following: (2)

- glycolysis
- the Krebs cycle
- oxidative phosphorylation

(ii) Any ***two*** of the following answers associated with those given in (i): (2)

- In glycolysis, energy is released in the oxidation of triose phosphate to produce ATP.
- In the Krebs cycle, ATP is produced in the regeneration of the 4-C compound from the 6-C compound
- In oxidative phosphorylation, energy is released to form ATP when electrons from NADH and FADH are passed from one carrier to the next.

(d) The oxygen produced in photosynthesis can be used in respiration. (1)

(7 marks)

##

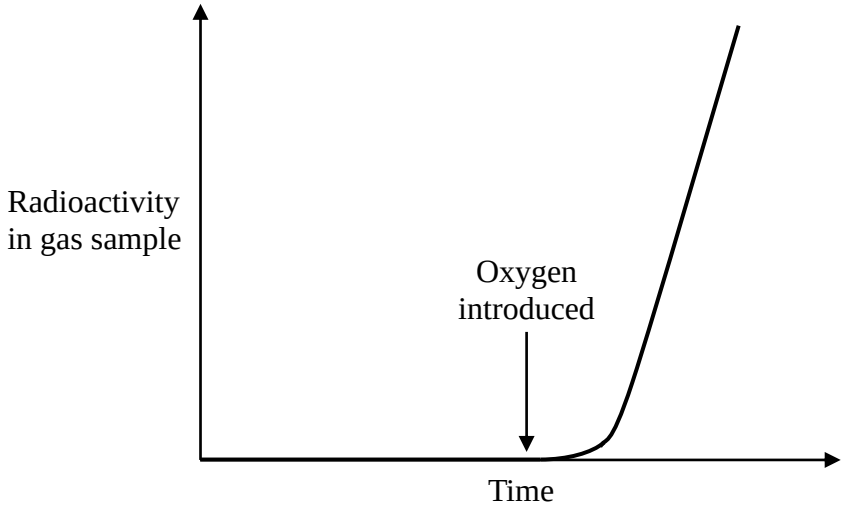
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	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Understanding basic concepts	1
(c)(i)	Understanding basic concepts	2
(c)(ii)	Applying concepts	2
(d)	Applying concepts	1

|!|BCESA0220000005|!| ★★ **SI**

In an investigation, a culture medium containing glucose labelled with radioactive carbon atoms was placed in a conical flask. A sample of animal cells was added to this medium.

The initial conditions in the flask were anaerobic. Oxygen was then bubbled through the medium after a period of time has passed. Samples of gas produced by the cells were tested for radioactivity at regular intervals. The graph shows the results.



(a) Explain why radioactivity only began to appear in the gas produced by the cells after oxygen was introduced. (3 marks)

(b) How would the results have differed if yeast cells were used in this investigation? Explain. (2 marks)

##

(a) Under the initial anaerobic conditions, radioactively-labelled glucose was broken down by glycolysis into radioactively-labelled pyruvate, which is not released from the animal cells. (1)
After oxygen was introduced, the radioactively-labelled pyruvate was then broken down in the link reaction / Krebs cycle. (1)
This produced radioactively-labelled carbon dioxide that the cell released as gas. (1)

(b) Radioactivity would have been found in the gas produced by the cells from the start of the experiment. (1)
This is because the yeast cells would have carried out alcoholic fermentation in the absence of oxygen, which would have released radioactively-labelled carbon dioxide gas. (1)

(5 marks)

##

	Skills assessed	Marks
(a)	Analysing data	3
(b)	Applying concepts	2

|!|BCESC0220000006|!|

When a person walks, most of the energy needed is provided by aerobic respiration. When a person runs a 100 m race, most of the energy needed is provided by anaerobic respiration.

(a) Explain why most of the energy needed for a 100 m race is provided by anaerobic respiration. (2 marks)

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(b) The person continues to breathe deeply for a while after the race. Explain why this is necessary. (2 marks)

##

(a) When exercising vigorously, the muscles cannot obtain enough oxygen to adequately carry out aerobic respiration in order to meet its energy demand. (1)
Under such circumstances, the muscles get the additional energy needed from anaerobic respiration. (1)

(b) Anaerobic respiration creates a toxic buildup of lactic acid in the muscles that must be removed. (1)
Breaking down lactic acid requires an extra amount of oxygen. (1)

(4 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	2

|!|BCESB0220000007|!| ★

The following flowchart shows the process of anaerobic respiration in yeast.

glucose (6-C) ⟶ 2 pyruvate (3-C) ⟶ 2 ethanol (2-C)

(a) Name this type of anaerobic respiration. (1 mark)

(b) Explain why the conversion of pyruvate to ethanol is essential for the continuation of this anaerobic respiration. (2 marks)

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(c) Give ***two*** industrial applications of this anaerobic respiration in yeast. (2 marks)

##

(a) Alcoholic fermentation (1)

(b) The conversion of pyruvate to ethanol regenerates NAD molecules. (1)
The NAD molecules can accept hydrogen in glycolysis again, thus allowing glycolysis to occur continuously. (1)

(c) Wine-making (1)
Bread-making (1)
(Or other reasonable answers)

(5 marks)

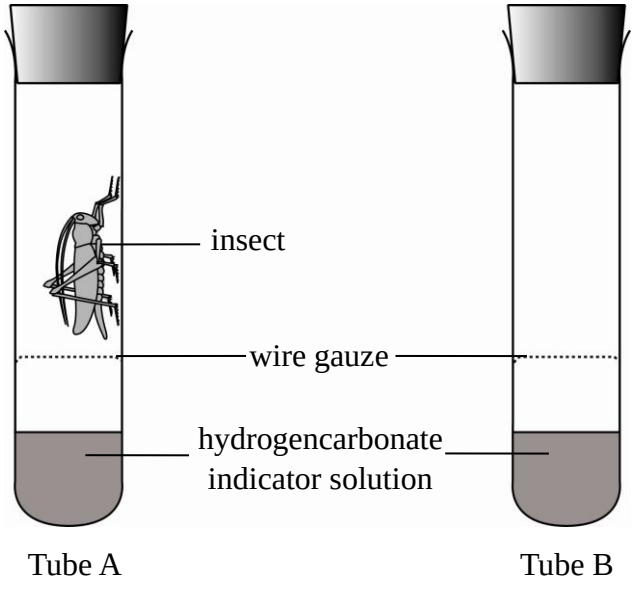
##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Applying concepts	2
(c)	Understanding basic concepts	2

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|!|BCESB0220000008|!| ★ **SI**

The diagram below shows the set-up of an experiment that demonstrated the respiration of an insect.



(a) Air was pumped into the hydrogencarbonate indicator solution for at least ten minutes before it was used in this experiment. What was the purpose of this procedure? (1 mark)

(b) Suggest ***two*** functions that tube B served in this experiment. (2 marks)

(c) The insect was kept in tube A for 30 minutes. What would be the colours of the indicator solutions in tubes A and B at the end of the experiment? (2 marks)

##

(a) This ensures that the carbon dioxide concentration in the indicator solution was equal to that of the air. (1)

(b) It acted as a control to show that the colour change of the indicator solution in tube A was due to the activity of the insect. (1)

It acted as a standard to which the colour of the indicator solution in tube A was compared with. (1)

(c) Tube A: yellow (1)

Tube B: red (1)

(5 marks)

##

	Skills assessed	Marks
(a)	Practical skills	1
(b)	Practical skills	2
(c)	Analysing data	2

|!|BCESC0220000009|!|

Susan made a loaf of bread using the following procedure.

(1) She mixed flour, yeast, sugar and warm water together to form dough.
(2) She put the dough in a warm place for about an hour.
(3) She baked the dough, which became a loaf of spongy bread.

(a) Explain why the bread would not taste alcoholic even though alcohol was produced in alcoholic fermentation. (1 mark)

(b) What would have been the result if Susan had not left the dough in a warm place for an hour? Explain why. (2 marks)

(c) Susan made another loaf of bread using the procedure above, except she placed the dough in a refrigerator instead of a warm place for an hour before baking. Explain why the dough would not rise. (2 marks)

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##

(a) The alcohol evaporated when the dough was baked. (1)

(b) The dough would not have risen. (1)

There would have been insufficient time for the yeast to ferment the sugar and produce carbon dioxide. (1)

(c) The rate of alcoholic fermentation in the yeast was low at low temperature. (1)

Thus insufficient carbon dioxide was produced to make the dough rise. (1)

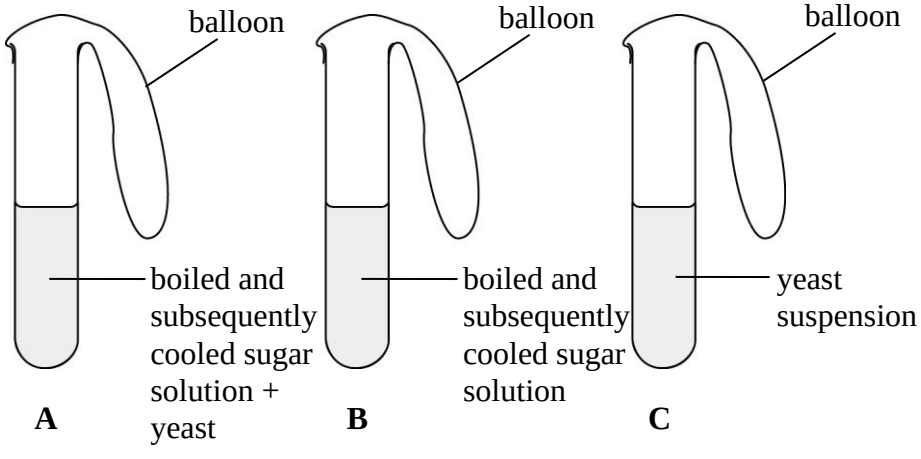
(5 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Applying concepts	2
(c)	Applying concepts	2

|!|BCESB0220000010|!| ★ SI

The following diagram shows the three set-ups used to investigate the activity of yeast. The set-ups were kept at 35 °C for two hours.



(a) Which set-up(s) would have inflated balloon(s) at the end of the experiment?

(1 mark)

(b) The gas in the inflated balloon was collected.

(i) What was the gas collected? (1 mark)

_____

(ii) How would you test for the gas? (1 mark)

_____

_____

(c) Which set-up(s) served as control(s)? (1 mark)

_____

_____

##

(a) Set-up A (1)

(b) (i) Carbon dioxide (1)

(ii) Carbon dioxide turns lime water milky OR turns hydrogencarbonate indicator solution yellow. (1)

(c) Set-ups B and C were controls. (1)

(4 marks)

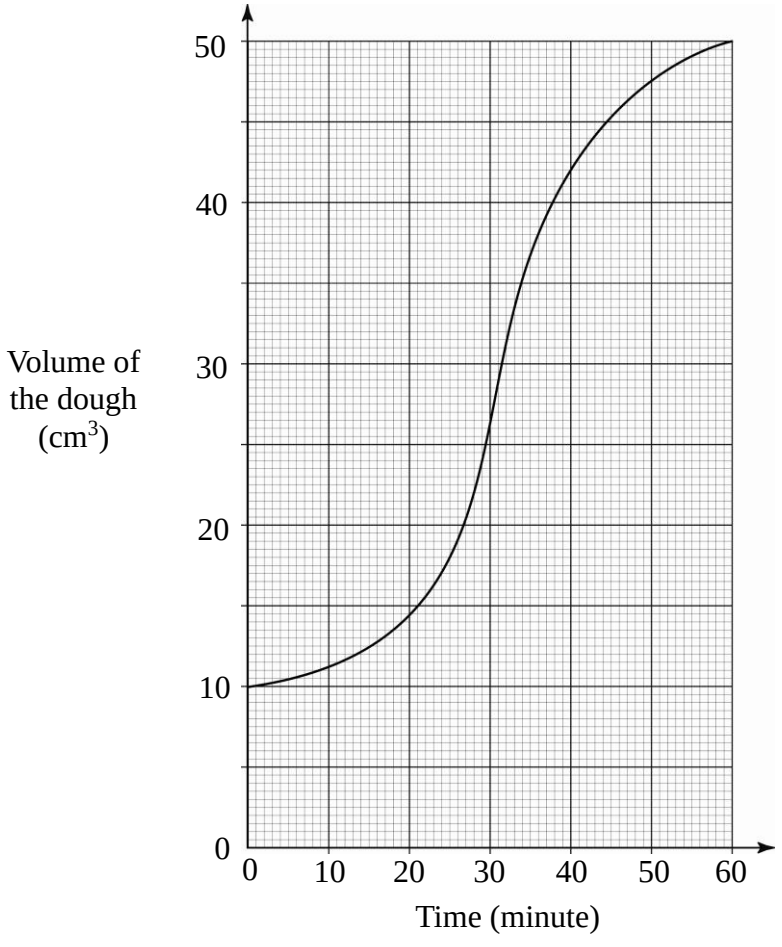
##

	Skills assessed	Marks
(a)	Analysing data	1
(b)(i)	Applying concepts	1
(b)(ii)	Practical skills	1
(c)	Practical skills	1

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|!|BCESB0220000011|!| ★ **SI**

In an investigation, yeast was mixed with water, sugar and flour to make dough, which was then kept at a constant temperature of 50 ºC in an oven. The volume of the dough was measured every five minutes. The results are shown in the following graph.



(a) From the graph, calculate

(i) the increase in the volume of the dough during the first 20 minutes of the investigation. (1 mark)

(ii) the percentage increase in the volume of the dough between 40 and 50 minutes. Show your working. (2 marks)

(b) Explain why there was an increase in the volume of the dough. (3 marks)

##

(a) (i) $(14.5 - 10)\ cm^3 = 4.5\ cm^3$ (1)

(ii) $\frac{47.5 - 42}{42} \times 100\%$ (1)

$= 13\%$ (1)

(b) The yeast carried out anaerobic respiration / alcoholic fermentation (1)
and produced carbon dioxide gas, (1)
which expanded upon heating. (1)

(6 marks)

##

	Skills assessed	Marks
(a)(i)	Mathematical skills	1
(a)(ii)	Mathematical skills	2
(b)	Applying concepts	3

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